DATA ENGINEER

LEARNING ROADMAP

Azure • Databricks • Snowflake • Airflow

A Structured Path to Modern Data Engineering

Table of Contents

Objective 3
Phase 1: Programming Fundamentals 4
1. Python 4
2. SQL 4
3. PySpark 4
Phase 2: Azure Data Engineering 6
Azure Data Factory (ADF) 6
Phase 3: Azure Databricks 7
Phase 4: Data Migration 8
SQL Server to Snowflake 8
Phase 5: Snowflake 9
Phase 6: Apache Airflow 10
Phase 7: End-to-End Project 11
Reference Architecture / Workflow 11
Best Practices 11
Capstone Project 11

Objective

This roadmap provides a structured learning path for becoming a Data Engineer with expertise in Azure, Databricks, Snowflake, and modern data engineering practices. It is organized into seven progressive phases, moving from programming fundamentals through cloud pipelines, orchestration, and a full end-to-end capstone project.

> **How to use this roadmap**
> Work through the phases in order — each builds on the previous one.
> Treat every topic list as a checklist to master before advancing.
> Reinforce learning by applying each phase to the end-to-end project in Phase 7.

Phase 1: Programming Fundamentals

Build the core coding foundation every data engineering task depends on.

1. Python

Learn the core concepts required for data engineering.

- Variables and Data Types
- Operators
- Conditional Statements
- Loops
- Functions
- Object-Oriented Programming (OOP)
- Exception Handling
- File Handling
- Modules and Packages
- Virtual Environments
- Working with APIs
- JSON Processing
- Logging
- Python Best Practices

2. SQL

Master SQL for data analysis and ETL development.

Area	Topics
Querying	SELECT Statements, Filtering Data, Sorting
Aggregation	Aggregate Functions, GROUP BY, HAVING
Joins	Inner Join, Left Join, Right Join, Full Join
Advanced	Subqueries, Common Table Expressions (CTEs), Window Functions, Views, Stored Procedures
Performance	Indexes, Performance Optimization

3. PySpark

Learn distributed data processing using Apache Spark.

- Spark Architecture
- DataFrames
- RDD Basics
- Reading Files
- Writing Files
- Transformations

- Actions
- Spark SQL
- Joins
- Aggregations
- Window Functions
- Partitioning
- Caching
- Performance Optimization
- Error Handling

Phase 2: Azure Data Engineering

Azure Data Factory (ADF)

Ingest and transform data at scale with Azure's cloud-native ETL/ELT service.

Area	Topics
Core Concepts	Integration Runtime, Linked Services, Datasets, Pipelines, Activities
Parameterization	Parameters, Variables, Expressions
Orchestration	Pipeline Triggers, Scheduling, Event Triggers
Operations	Debugging, Monitoring, Logging
Mapping Data Flows	Derived Column, Conditional Split, Filter, Lookup, Join, Aggregate, Pivot / Unpivot, Flatten
Dimensional	Surrogate Key Generation, Slowly Changing Dimension (SCD)
Loading	Copy Activity, Incremental Load, CDC (Change Data Capture)

Phase 3: Azure Databricks

Develop and optimize Spark workloads, Delta Lake, and data-quality pipelines.

Area	Topics
Core Concepts	Workspace, Clusters, Notebooks, Jobs, Unity Catalog, DBFS, Secrets Management, Widgets, Libraries, Git Integration
PySpark Development	DataFrame API, Spark SQL, UDFs, Optimization, Caching, Partitioning, Broadcast Joins
Delta Lake	ACID Transactions, Time Travel, Schema Evolution, Merge Operations, Upserts, Deletes, Optimize, Vacuum, Z-Ordering
Delta Live Tables (DLT)	Live Tables, Streaming Tables, Materialized Views, Expectations, Pipeline Configuration, Incremental Processing, Error Handling, Monitoring, Pipeline Scheduling
Data Quality Checks	Null Validation, Duplicate Detection, Primary Key Validation, Referential Integrity, Schema Validation, Business Rule Validation, Data Freshness Checks, Record Count Validation, Custom Expectations, Error Logging

Phase 4: Data Migration

SQL Server to Snowflake

Plan and execute a reliable migration from a SQL Server source into Snowflake.

- Source System Analysis
- Schema Migration
- Data Type Mapping
- Initial Data Load
- Incremental Load
- CDC
- Bulk Loading
- Snowpipe
- Error Handling
- Performance Tuning
- Validation
- Reconciliation

Phase 5: Snowflake

Master the cloud data warehouse that anchors the analytics layer.

Area	Topics
Architecture & Objects	Architecture, Databases, Schemas, Tables, Virtual Warehouses
Staging & Loading	Stages, File Formats, Internal & External Stages, COPY INTO, Snowpipe
Automation	Streams, Tasks
Optimization & Storage	Clustering, Time Travel, Zero Copy Clone, Materialized Views, Performance Optimization
Sharing & Security	Secure Views, Data Sharing, Security, RBAC

Phase 6: Apache Airflow

Orchestrate, schedule, and monitor end-to-end data pipelines.

Area	Topics
Core Concepts	DAGs, Operators, Sensors, Hooks, Scheduling, Task Dependencies
State & Flow	Variables, Connections, XCom, Branching, Dynamic DAGs, Retry Policies
Operations	Error Handling, Logging, Monitoring
Orchestration	Workflow Orchestration, ETL Scheduling, Pipeline Dependencies, Backfilling, Incremental Pipelines
Alerting	Failure Notifications, Email Alerts, SLA Monitoring

Phase 7: End-to-End Project

Build an enterprise-grade Azure Data Engineering project that ties every phase together.

Reference Architecture / Workflow



Best Practices

- Follow Medallion Architecture (Bronze, Silver, Gold)
- Implement CI/CD using Azure DevOps or GitHub
- Use Git Version Control
- Parameterize Pipelines
- Build Reusable Frameworks
- Implement Logging
- Monitor Pipelines
- Secure Secrets using Azure Key Vault
- Optimize Spark Jobs
- Optimize Snowflake Warehouses
- Implement Data Validation at Every Layer

Capstone Project

Develop a complete data engineering solution that includes:

- SQL Server as the source

- Azure Data Factory for ingestion
- Azure Data Lake Storage
- Azure Databricks with PySpark
- Delta Live Tables
- Data Quality Framework
- Snowflake Data Warehouse
- Apache Airflow for orchestration
- Power BI for reporting

Outcome

This project demonstrates a complete end-to-end modern data engineering pipeline using Azure and Snowflake technologies — from source ingestion through orchestration and reporting.